

Workshop Activities

Activity 1 – Java Cup Taste Data

Huron Lakes Candies (HLC) has developed a new candy bar called Java Cup that is a milk chocolate cup with a coffee-cream centre. In order to assess the market potential of Java Cup, HLC has developed a taste test and follow-up survey. Respondents were asked to taste Java Cup and then rate Java Cup's taste, texture, creaminess of filling, sweetness, and depth of the chocolate flavour of the cup on a 100-point scale. The taste test and survey were administered to 217 randomly selected adult consumers. Data collected from each respondent are provided in the file *JavaCup*.

- a. Are there any missing values in HLC's survey data? If so, identify the respondents for which data are missing and indicate which values are missing for each of these respondents.
- b. Are there any values in HLC's survey data that appear to be erroneous? If so, identify the respondents for which data appear to be erroneous and indicate which values appear to be erroneous for each of these respondents.

Activity 2 – Bias

Consider these three studies:

1. In the 1930s, Dr. Joseph Rhine tested 500 participants to guess the order in a shuffled deck of cards. Only those participants who guessed correctly moved to the next round. The last participant to guess right every round was declared to have telepathic abilities.
2. A policy group lobbying for the alcohol industry conducted an exit poll outside of a popular restaurant late at night. The restaurant patrons were asked whether driving after drinking was a serious problem. The policy group concluded that driving after drinking was not a serious problem.
3. A research group followed for 40 years a cohort of individuals. The study's main conclusion was that those who drank red wine were richer, better educated, had lower levels of cholesterol ratio, and consequently a lower risk of heart attacks than those who did not.

Associate each of the three studies to one form of bias (Selection bias, Simpson's Paradox, Survivor Bias) and explain your choice.

Activity 3 – Amount of Vacation Used at Consulting Firm

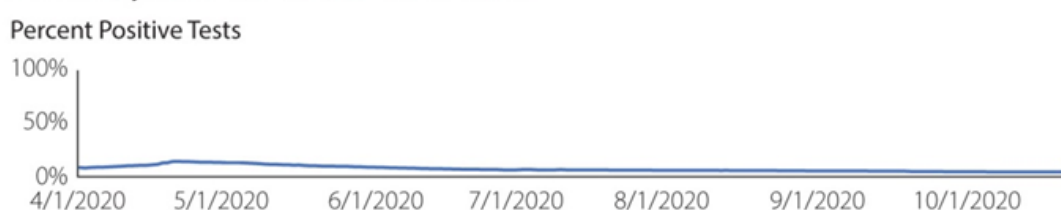
Guaraldi and Associates is a management consulting firm located in Manhattan. The firm is interested in examining the relationship between the amount of vacation that their consultants use per year and how long the consultant has been with the firm. All consultants at Guaraldi and Associates, other than the Managing Partners, are considered either Junior Consultants or Senior Consultants based on their skills and expertise. The file *GuaraldiConsultants* contains data on all Junior and Senior Consultants at Guaraldi and Associates including the years of service the consultant has at the firm and the number of hours that the consultant took as vacation last year.

- Create a scatter chart and add a trendline to examine the relationship between years of service and amount of vacation time used for all consultants at Guaraldi and Associates. Based on the scatter charts, what appears to be the relationship between years of service and amount of vacation time used?
- Create a scatter chart for the same data, but this time differentiate the points in the scatter chart based on whether the consultant is junior or senior. Do you see the same relationship in this scatter chart individually for junior and senior consultants as you saw in part a for all consultants?
- Do the results in parts a and b provide an example of Simpson's paradox? Discuss why or why not?

Activity 4 – COVID-19 Positivity Rates in Ohio

An epidemiologist has collected data on positivity rates for COVID-19 testing in the state of Ohio. The positivity rate is defined as the number of tests that come back positive for the COVID-19 virus divided by the total number of COVID-19 tests completed. The epidemiologist has created the following chart showing positivity rates in Ohio between April and October 2020. The audience for this chart is interested in trying to gain insights into any overall increasing or decreasing trends in the positivity rate for COVID-19 during this time period.

Positivity Rate for COVID-19 in Ohio



- a. How can this chart be improved to meet the needs of the audience?
- b. Use the data in the file *PositivityOhioChart* to redesign the chart to present a more effective data visualization for an audience trying to gain insights into the positivity rate of COVID-19 testing in Ohio. Justify your redesign choices.

Activity 5 – Headcount and Revenue at Maximus Fashion.

Maximus Fashion is a high-end clothing store in Naperville, Illinois. The store manager would like to create a data visualization that shows trends in the store's headcount of retail associates and revenues. For the last eight quarters, the manager has collected data on the store's headcount of retail associates (expressed in terms of full-time equivalent [FTE] employees) and the store's revenues. The manager has chosen to display these data as the dual-axis chart that follows. The audience for this chart is an informal advisory group for the store who has considerable experience in fashion, but little experience with financial analysis or data visualization in general.

- a. What possible confusion could the use of this chart create for the audience?
- b. Using the data in the file *MaximusFashionChart*, redesign this dual-axis chart to be more effective for the audience. Justify your redesign choices.

Activity 6 – Continue with Task 2

Please use the remaining time in this workshop to continue working on Task 2 which is due to be submitted this week.